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MILITARY SPECIFICATION MANUAL, TECHNICAL: TRANSPORT GUIDANCE FOR MILITARY MATERIEL, PREPARATION OF



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MILITARY SPECIFICATION

MANUAL, TECHNICAL:
TRANSPORT GUIDANCE
FOR MILITARY MATERIEL,
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This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

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1. SCOPE.

1.1 General. This specification covers the format, style, and content requirements for the preparation of Transport Guidance Technical Manuals (TGTMs). TGTMs provide guidance for the worldwide movement of military materiel.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.4).

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be useful in improving this document should be addressed to: Director, MTMC Transportation Engineering Agency, ATTN: MTTE-TR, Box 6276, Newport News, VA 23606-0276, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY SPECIFICATIONS:

MIL-S-22824 Slings, Multiple Leg, Vehicle (Shipboard Loading)
MIL-M-38784 Manuals, Technical: General Style and Format Requirements

FEDERAL STANDARD:

FED-STD-376 Preferred Metric Units for General Use by the Federal Government

MILITARY STANDARDS:

MIL-STD-12 Abbreviations for Use on Drawings, and In Specifications, Standards and Technical Documents
MIL-STD-209 Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
DOD-STD-00100 Engineering Drawing Practices
MIL-STD-1366 Materiel Transportation System Dimensional and Weight Constraints, Definition of
MIL-STD-1791 Designing for Internal Aerial Delivery in Fixed Wing Aircraft
AR 25-30 The Army Integrated Publishing and Printing Program
AR 310-50 Authorized Abbreviations
DOD 6050.5-L Hazardous Material Information System (HMIS)
(Superintendent of Documents, US Government Printing Office, Washington, DC 20402)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. If a conflict occurs between the guidance in this document and references cited herein (except for related detail specifications, specification sheets, or MS standards), the guidance in this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 General. The TGTM shall contain all essential information required by personnel engaged in the various phases of worldwide transport. Emphasis should be on guidance. While every effort shall be made to describe the most likely transport method, alternatives should be offered.

3.1.1 Format and manuscripts. Detailed information on style and format and the preparation of manuscripts and reproducible copy of technical manuals is in MIL-M-38784.

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3.1.2 Reading level. The reading grade level shall be the level specified by the contracting activity (see 6.4). MIL-M-38784 shall be used to determine the TGTM reading grade level.

3.1.3 Style. The text shall be readily understandable to a person with a minimum amount of training in transportation. The text shall be factual, concise, and in the simplest English that will convey the intended message. Technical phraseology requiring special knowledge shall be avoided except where no other wording will convey the meaning. Every effort shall be made to avoid confusing the reader with lengthy paragraphs or verbiage (see MIL-M-38784).

3.1.4 Selection and use of illustrations. Illustrations shall be used to support text as necessary for identification, emphasis, clarification, and visual demonstration. Illustrations shall not be prepared solely to appeal to reader interest, for decorative purposes, or to show operations or objects of basic knowledge. Progressive step-type or exploded-view illustrations, or a combination of both, shall be used when practical in place of lengthy narrative. If equipment to be illustrated is subject to frequent engineering changes, it is permissible to use one figure that presents a typical view of the subject. The title of such figures shall contain the word "typical."

3.1.4.1 Illustration composition. In composing the parts of an illustration, the handling of the subject depends on the purpose of the illustration. The illustration's subject shall be displayed to the best advantage consistent with the illustration's purpose and the limitations of the drawing methods used. If possible, illustration subjects shall be displayed from a normal viewing position. A drawing that views an isolated part from an unnatural angle may confuse the reader. When it is necessary to show a detail from an unnatural position, use some type of orientation or identification view.

3.1.4.2 Foldouts. Unless otherwise specified (see 6.4) foldout pages and illustrations over one page in size shall not be used.

3.2 Content. Unless otherwise specified (see 6.4), content and order of presentation shall be as follows:

	Para No.
FRONT MATTER.	(3.3)
CHAPTER 1 - Introduction. a. Section I - General. b. Section II - Safety. c. Section III - Equipment Description. d. Section IV - Containerization.	(3.4)

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CHAPTER 2 - Highway Transport. (3.5)
a. Section I - General.
b. Section II - Self-Delivery.
c. Section III - Motor Vehicle Transport.
d. Section IV - Containerization.

CHAPTER 3 - Rail Transport. (3.6)
a. Section I - General.
b. Section II - Rail Loading.
c. Section III - Containerization.

CHAPTER 4 - Marine Transport. (3.7)
a. Section I - General.
b. Section II - Ship Loading.
c. Section III - Containerization.

CHAPTER 5 - Air Transport. (3.8)
a. Section I - General.
b. Section II - Transport by Cargo Aircraft.

References and Appendixes. (3.9)

3.3 Front matter. Front matter shall consist of a cover/title page, warning page (see 3.3.2), and a reporting errors/table of contents page.

3.3.1 Cover/title page. The cover/title page shall be as shown in figure 1 and include the following:

a. Publication number: TM 55- (assigned by US Army Materiel Command)

b. "TRANSPORT GUIDANCE TECHNICAL MANUAL"

c. Representative line drawing of item (if space permits).

d. Item nomenclature (see 3.4.3.5)/national stock number (NSN) if applicable; and if TGTM covers only one item). Use back of front cover for equipment lists and NSNs.

e. Appropriate distribution statement, destruction notice, and export control warning notice (if applicable).

f. Publication date - month and year.

3.3.2 Warning page. When specified (see 6.4), a general-type warning page shall be included in the TGTM (see MIL-M-38784).

3.3.3 Reporting errors/table of contents (see MIL-M-38784).

3.4 CHAPTER 1 - Introduction. This chapter and each subsequent chapter shall begin on a right-hand page. It shall contain:

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	Para No.
a. Section I - General.	(3.4.1)
b. Section II - Safety.	(3.4.2)
c. Section III - Equipment description.	(3.4.3)
d. Section IV - Containerization.	(3.4.4)

3.4.1 Section I - General. This section shall contain the following:

	Para No.
a. Purpose and scope.	(3.4.1.1)
b. Related publications.	(3.4.1.2)
c. Definitions.	(3.4.1.3)
d. Warnings.	(3.4.1.4)
e. Responsibilities and tasks.	(3.4.1.5)

3.4.1.1 Purpose and scope. This paragraph shall state why the manual was prepared, what areas it covers, who shall use it, and when it is to be applied.

3.4.1.2 Related publications. This paragraph shall contain reference to publications that can supplement the material presented in a given TGTM chapter. Examples are:

- Railcar Loading Procedures, TM 55-601.
- Strategic Deployment by Surface Transportation, FM 55-65.
- Army Helicopter External Load Operations, FM 55-450-1.
- Army Helicopter Internal Load Operations, FM 55-450-2.
- Application of Blocking, Bracing, and Tiedown Materials for Rail Transport, TM 55-2200-001-12.
- Preparation of Hazardous Materials for Military Air Shipment, TM 38-250/AFR 71-4/NAVSUP PUB 505(Rev)/ MCO P4030.19D/DSAM 4145.3.

3.4.1.3 Definitions. This paragraph shall contain definitions of general terms or acronyms used in most TGTMs. Examples are loading restraint factors (LRFs), curb weight (CW), vehicle payload, and gross vehicle weight (GVW).

3.4.1.4 Warnings. Warnings, cautions, and notes are means of attracting attention to essential or critical information in the TGTM. Examples and method of presentation are shown in MIL-M-38784.

WARNING

Conditions, practices, or procedures that must be observed to avoid:

a. Personal injury.
b. Loss of life.

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CAUTION

Conditions, practices, or procedures that must be observed to avoid:

a. Damage to equipment.
b. Destruction of equipment.
c. Long-term health hazards.

NOTE

Information of special:
a. Importance.
b. Interest.
c. Aid in job performance, which-
   (1) Should be remembered.
   (2) Will facilitate decision.
   (3) Would otherwise be difficult to find.
   (4) Would otherwise be difficult to incorporate.

3.4.1.5 Responsibilities and tasks. This paragraph identifies responsibilities and tasks to be performed.

Example: For the installation transportation officer (ITO), this statement shall appear in the TGTM: "The installation transportation officer (ITO) issuing the bill of lading for shipment of the item is responsible for determining the appropriate rail and motor freight classification."

3.4.2 Section II - Safety. This section shall identify general safety considerations and precautions relative to transportation. Hazardous characteristics (see 3.4.2.1) that would endanger personnel, material, carrier vehicle, or military or civilian communities shall be emphasized by a warning (3.4.1.4). Pertinent safety requirements for an individual transport mode shall be identified in order of severity in the relevant chapter.

3.4.2.1 Hazardous material considerations. Present the following information, as applicable, for each item classified as hazardous material:

- US Department of Transportation (DOT) hazard class.
- Proper shipping name and label requirements.
- Placarding requirements.
- Disaster response force requirements.
  -- Protective clothing.
  -- Medical.
  -- Firefighting.
- Security.
- Any other pertinent information, such as:
  -- Venting or grounding requirements.
  -- Storage compatibility codes.

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(Portions of the above-listed information and specific Material Safety Data Sheet (MSDS) information may be obtained from DoD 6050.5-L, Hazardous Material Information System (HMIS).)

3.4.3 Section III - Equipment description. This section shall describe the equipment and contain a listing of administrative and technical data that influence an item's transportability. When the subject system contains several distinct items, the TGTM will cover each vehicle, trailer, or other component separately. The following shall be included:

	Para No.
a. General description.	(3.4.3.1)
b. Administrative and technical data.	(3.4.3.3 and 3.4.3.4)
c. Technical drawings.	(3.4.3.6)

3.4.3.1 General description. This paragraph shall introduce the reader to the equipment and its military nomenclature (see 3.4.3.5). It shall include a brief written description of the item and its intended use, as well as an identifying illustration (see 3.4.3.2). Differences in models, and other pertinent information such as manufacturer and commercial designation, shall be provided. The reader shall be referred to operator and unit maintenance manuals and other publications listed in the appendix for additional information.

3.4.3.2 Equipment identification. The first figure in chapter 1 shall be an equipment-identifying illustration. All TGTM illustrations shall be line drawings. The identifying drawing shall be sized to cover at least one-half of a printed page. It should be a three-quarter front or rear view of the item in its operational configuration. When a TGTM covers more than one model of equipment and if, for transport purposes, the equipment is almost identical, only one illustration is required. Line drawings of different models or system components shall be included only when they exhibit significant external differences.

3.4.3.3 Administrative data. The following information shall be presented in tabular form for each item covered in the TGTM.

- Nomenclature of each item.
- National stock number (NSN).
- Line item number (LIN).

3.4.3.4 Technical data. This information shall be combined with the administrative data (3.4.3.3) and presented as shown in figure 2. Unless otherwise noted, technical data shall apply to an operational configuration. If the equipment should be "reduced" or "sectionalized" for transport (see 3.4.3.8) reduced dimensions shall also be given.

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a. Overall dimensions (see 3.4.3.9).

- Length, width, height (height at both CW and GVW if change is significant), and resulting area and volume.

b. Lifting and tiedown provisions (see 3.4.3.7).

-Location.

c. Weight (see 3.4.3.9).

- Curb weight (CW).
- Gross vehicle weight (GVW).
- Payload (GVW less CW).
  -- Identify allowance for crew weight.

- Weight distribution at GVW (and CW when appropriate):
  -- On each axle.
  -- On kingpin (or tractor 5th wheel), or trailer landing gear, or blade, bucket, and gun-carriage trail.
  -- On road wheels of tracked vehicles.
  -- On each skid.

d. Center of gravity (CG) at GVW (and CW when appropriate).

- Distance above ground and reference base line.
  -- From centerline of front axle; or drive sprocket; or kingpin; or some other clearly defined reference point.
  -- When CG is not on the longitudinal axis; give offset.

The applicable portions of the following data (when available) shall be included in this general paragraph:

e. Towing capacity: Allowable trailer GVW and effect on payload (see 3.5.3.1).

f. Tire information: Number (quantity), size, ply rating, and inflation pressure.

g. Track information: Type of track, and actual ground contact area.

h. Footprint data: Ground pressure resulting from track, tire (or group of tires as appropriate), and skid contact at GVW (operational weight).

i. Highway speed and range: Primarily for self-propelled, self-delivered vehicles.

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j. Turning radius: See 3.5.3.2.

k. Ground clearance: Normally lowest point under differential or suspension component.

l. Ramp clearance: Expressed as the maximum ramp angle between two horizontal surfaces that a vehicle can pass without interference at the ramp foot or crest.

m. Item weight crated and uncrated: Primarily for non-vehicular equipment.

3.4.3.5 Nomenclature. Descriptive nomenclature of equipment, once named in the manual, may be shortened to avoid needless repetition, provided retained descriptive nomenclature ensures clear identification. For example, high-mobility multipurpose wheeled vehicles, M998-series, may be expressed as M998, or HMMWV.

3.4.3.6 Technical drawings. Side and end elevation drawings (see 3.4.3.10 and 3.4.3.11) shall show the item in its reduced (sectionalized) configuration, with removable portions of the operational configuration indicated by phantom lines. Figure 2 shows typical format and content. Drawings shall be prepared in accordance with those general practices outlined in DOD-STD-00100. Abbreviations shall be in accordance with MIL-STD-12 for drawings and AR 310-50 for text.

3.4.3.7 Lifting, tiedown, and cargo provisions. The following information shall be provided as indicated.

a. Identifier. Lifting and tiedown provisions, tow points (pintle or lunette), and tiedown hard points shall be identified on the technical drawings (3.4.3.6). The same feature designator (number and identifier mark) shall be used throughout the TGTM. In this way, a lifting provision shown in chapter 1 will be readily identifiable in successive chapters. Number each provision consecutively from front to rear starting on the roadside.

b. Rating. Numeric strength values shall not be given for lifting and tiedown provisions. Provisions on new equipment will be designed and tested IAW MIL-STD-209. TGTM procedures shall take into consideration provision capacity, location, and expected lifting or restraining forces. Figures 2 and 3 show examples of vehicle data.

c. Cargo. The load ratings for cargo tiedowns shall be as defined by MIL-STD-209.

3.4.3.8 Reduced and Sectionalized. These terms are defined as:

a. "Reduced" denotes removal, retraction, or folding of vehicle or item components to minimize shipment profile. While

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varying degrees of reduction are possible, dimensional reduction of vehicles for logistical moves is generally limited to measures that can be readily performed at the organizational level and do not impair the mobility of the vehicle. Reduction may include folding down windshield, folding back or removing side mirrors, and/or removing such items as exhaust-pipe extension, cargo body tarpaulin and bows, side racks, spare tire or wheel, or similar protruding parts. Removed components must be packaged or bundled and shipped in or near the vehicle from which they were removed.

b. "Sectionalized" refers to physically separating major components to reduce both size and weight of an item's shipping configuration. Vehicle sectionalization for logistical moves is generally limited to measures that can be performed at the organizational level. Materials handling equipment (cranes or forklift trucks) may be required. Sectionalization will, in most cases, impair vehicle mobility.

3.4.3.9 Measurements and tolerances. Numerical data shall be presented in both the International System of Units (SI) and US Customary units. (Refer to FED-STD-376, Preferred Metric Units for General Use by the Federal Government.) Unless otherwise specified (see 6.4), TGTMs will use customary (inch-pound) units, with the SI (metric) equivalent shown in parentheses. Generally, only major dimensions or measurements, such as overall size and weight, need be expressed in both systems.

a. Equipment dimensions shall be to the nearest whole inch, unless the dimension is known to be valid to ·1/4 inch. Such dimensions will be rounded to one decimal place.

Example:

Width of 86-3/8 in. (valid to 1/8 in.) = 86.375 in.
Shown in TGTM as 86.4 in.
Height of 110-1/4 in. (about) = 110.250
Shown in TGTM as 110 in. (or 111 in. if judged appropriate to account for variations in vehicle suspension and tire deflection).

b. Derived data shall be rounded to nearest whole 5 or 10 (depending on the magnitude of the number) within 2 percent of the mathematic value.

Example:

Area: 86.375 X 221 ÷ 144 = 132.562 ft^2
132.56 X 102 pct = 135.21
Shown in TGTM as 135 ft^2

Cube: 86.375 X 110.25 X 221 ÷ 1,728 = 1,217.910 ft^3
1,217.91 X 102 pct = 1,242.27
Shown in TGTM as 1,220 ft^3

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c. Conversions (US Customary to SI, or the reverse) shall be as rational as practical. When the value is known to be valid to a specific tolerance, the converted value shall be given to the same tolerance.

Example of toleranced conversion:
Width of 86-3/8 in. (valid to 1/8 in. or about 3 mm)
86.375 X 25.4 = 2,193.9 mm
2193.9 + 3 = 2196.9
2193.9 - 3 = 2190.9
Shown in TGTM as 219 cm (or 2190 mm)

d. Untoleranced inch-pound values shall be converted to the nearest whole number metric unit within 2 percent of the mathematic value. Generally, the last digit (rightmost) of untoleranced conversions can be further rounded to the nearest 5 or 0.

Example of untoleranced conversion:
Tire pressure: 28 psi X 6.895 = 193.060 kPA
193.06 X 98 pct = 189.20 (-2 pct)
193.06 X 102 pct = 196.92 (+2 pct)
Shown in TGTM as 190 kPa
Turn radius (see 3.5.5): 20 ft X 0.3048 = 6.096 m
6.096 X 98 pct = 5.974
6.096 X 102 pct = 6.218
Shown in TGTM as 6 m
Cube: 1,220 ft^3 X 0.0283 = 34.526 m^3
34.53 X 98 pct = 33.84
34.53 X 102 pct = 35.22
Shown in TGTM as 35 m^3

3.4.3.10 Side elevation. Side elevation drawings shall show the left side (driver's left side) of a vehicle or trailer (figure 2). Either side of nonvehicular equipment may be shown, but showing the side with the greater number of projections or other irregularities that would make securing of the item difficult is preferable. Side elevation drawings shall show the location of lifting and tiedown provisions and give the overall height and length dimensions. The following dimensions, if applicable, shall also be shown:

a. Horizontal dimensions.

(1) Distance between centerlines of axles.

(2) Distance from each end to centerline of nearest axle.

(3) Overall reduced length. Annotate "reducible to."

(4) Length of ground contact for tracked items. Annotate "ground contact length" immediately below the dimension line.

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(5) Center of gravity (3.4.3.4).

(6) Distance between lifting provisions (and to vehicle centerline or clearly defined equipment reference).

(7) Distance between tiedown provisions (same as above).

(8) Location and size of cargo bed and cargo tiedown fittings.

(9) Skid ground-contact dimensions.

(10) Other applicable or special dimensions, such as pintle location, significant profile changes, locations of kingpin and landing gear on semitrailers, or centerline of bogie assemblies or drive sprocket.

b. Angular dimensions.

(1) Angle of approach and angle of departure.

(2) Ramp crest angle, to determine clearance between ramp crest and underside of vehicle (generally long wheelbase).

c. Vertical dimensions.

(1) Center of gravity (3.4.3.4).

(2) Ground clearance. Show location and distance from lowest part of undercarriage to ground level.

(3) Overall operational and reduced heights. Annotate "reducible to."

(4) Height above ground to lifting provisions.

(5) Height above ground to tiedown provisions.

(6) Other applicable or special dimensions, such as significant profile changes, or height to lunette or gooseneck of semitrailer.

3.4.3.11 End elevation. The end elevation drawing that will best show the major projections or contours shall be included (figure 3). This drawing shall show the location of lifting, towing, and tiedown provisions and the overall width dimension. The following dimensions, if applicable, shall also be shown:

a. Distance between outside edges of skids, tracks, or wheels.

b. Distance between centerlines of left and right skids, tracks, wheels, or set of dual wheels.

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c. Width of one skid, track, or wheel or set of dual wheels.

d. Overall reduced width. Annotate "reducible to."

e. Ground clearance. Show height dimension between ground and lowest part of vehicle underbody.

f. Location of transverse center of gravity.

g. Other applicable or special dimensions, such as significant profile changes, height of towing pintle, truck bed, or fifth wheel, or width of cab.

3.4.3.12 Scale. The side and end elevations shall not be to any specific scale but shall provide proper proportion and ready identification of the item they represent. Each view shall be displayed as large as possible on the printed page, if necessary, reduce or enlarge the original artwork. The side elevation of tracked and wheeled vehicles shall be broadside to the page; the end elevation may be either vertical or broadside to the page.

3.4.4 Section IV - Containerization. When specified in the TGTM contract (see 6.4), information on loading and tying down the item in ISO-type containers and MILVANs shall be presented.

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3.5 CHAPTER 2 - Highway transport. This chapter shall include the following:

		Para No.
a.	Section I - General.	(3.5.1)
b.	Section II - Self-delivery.	(3.5.3)
c.	Section III - Motor vehicle transport.	(3.5.4)
d.	Section IV - Containerization.	(3.5.5)

3.5.1 Section I - General. This section shall contain general information for highway movement. When the shipping configuration exceeds limitations of CONUS or overseas areas, a complete description shall be presented about regulatory controls. In TGTMs covering more than one item, the particular paragraph for each individual load shall identify the pertinent limitations. The TGTM shall clearly identify item characteristics (size or weight) that exceed CONUS or OCONUS limitations for permits or administrative approval. The TGTM shall reflect the limitations for CONUS or overseas areas, described in the MTMCTEA Transportability Engineering Analysis on the specific item.

3.5.2 Safety. Besides the general safety requirements of chapter 1, specific safety precautions peculiar to self-delivery or motor vehicle transport shall be identified in order of severity. Include applicable items listed in 3.4.2.1.

3.5.3 Section II - Self-delivery. For an item considered self-deliverable, the manual shall include a section for this method. It shall contain, but not be limited to, operating range, speed, and special considerations. An example of a special consideration is: "Although the tracks of the M1A1 tank have rubber pads, movement over paved public highways should not be attempted without approval from highway authorities."

3.5.3.1 Towing. Towing capacity and speed shall be indicated along with any reduction in vehicle payload caused by the vertical load on towing pintle (3.4.3.4e).

3.5.3.2 Turning radius and clearance. Turning radius and turning clearance are defined as:

a. Vehicle turning radius is the outermost radius traced by a vehicle's bumper or other projection with the steering at full lock. Folding side mirrors are not to be included in the vehicle turning radius or turning clearance.

b. Turning clearance is a measure of the space a vehicle (truck, tractor and semitrailer, or truck and full trailer) needs to make a 90° turn. When requested (see 6.4), turning clearance is given in terms of maximum lane width traced by a vehicle

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(including overhang) making a 90° turn. The turning maneuver begins with the steering at full lock. Runout on the 90° heading change begins when the inboard steer-wheel reaches the new heading.

3.5.4 Section III - Motor vehicle transport. This section shall contain guidance on hauling-vehicle (prime mover) selection, item preparation, loading, blocking and bracing, and tiedown.

3.5.4.1 Vehicle selection. Truck or trailer selection shall take into account appropriateness and availability. In general, the TGTM item weight should closely match the selected vehicle's payload and physical size. The selected vehicle should be in the same unit TOE or in a support unit that could be expected to provide transport. To aid planning, a listing shall be included of suitable trucks, trailers, and semitrailers. This listing will indicate which vehicle is or is not usable with the given item. Figure 4 shows sample format and content.

3.5.4.2 Preparation. This section shall describe item preparation peculiar to highway transport to include safety, economy, and measures to curtail pilferage en route. When practical, guidance shall identify specific reduction (3.4.3.8) required for unrestricted movement in CONUS and overseas areas. Unit load integrity shall be maintained insofar as possible; that is, any component or part removed should be placed within the cube of the basic item.

3.5.4.3 Special transport equipment. When applicable, the TGTM shall identify requirements for special shock and vibration-mitigating transport vehicles, such as those equipped with air-cushion or other special suspension systems.

3.5.4.4 Axle loads. Individual axle loads for the selected truck/trailer combination shall be included. For semitrailers, the kingpin load (static fifth wheel load) shall also be stated.

3.5.4.5 Load restraint. The loading restraint factors (LRFs) used for surface and air modes are the "g" (acceleration of gravity) loading factors expected in military transport. The LRFs used in the highway transport chapter shall be given in a TGTM appendix. Unless otherwise stated in the MTMCTEA Transportability Engineering Analysis, highway LRFs are:

0.7 in the forward direction (relative to the transporter).
0.3 in the aft and vertical directions.
0.1 in the lateral direction.

The tiedown arrangements shown in the TGTM shall be based on the following:

- restraint factors applied independently in each direction.

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- restraint load (GVW times the LRF) resolved into resultant lashing loads, allowing for tiedown angle.

- lashing loads less than safe working load (SWL) of the restraint (that is, chain).

3.5.4.6 Bill of materials. The bill of materials shall list the material nomenclature of each item; a description of the item, with size, grade, type, and applicable Federal Specification or Standard reference and NSN; and the required linear footage to restrain the item on the transport vehicle. Figure 5 shows sample format and content.

3.5.4.7 Blocking and tiedown procedures. Blocking and tiedown procedures shall present a loading arrangement using, where feasible, commercially available restraint devices and materials that will provide adequate highway loading restraint (3.5.4.5). Lifting and tiedown provisions shall retain the designation shown in chapter 1 (3.4.3.7). A table of instructions outlining the blocking and tiedown procedures shall accompany each tiedown diagram. Figure 6 shows sample format and content. The table shall be indexed to the blocking and tiedown diagram and shall identify the quantity required and steps necessary to ensure proper installation. Highway tiedown information should be in the same general format as used for rail. Figure 7 shows a sample rail format. Where general instructions apply, they shall be listed at the bottom of the table.

3.5.5 Section IV - Containerization. When applicable, this section shall provide procedures for shipment of equipment in MILVANs or commercial containers. Procedures shall include item preparation and required blocking, bracing, and tiedown.

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3.6 CHAPTER 3 - Rail transport. This chapter shall provide procedures for safe, economic railroad shipment.

	Para No.
a. Section I - General.	(3.6.1)
b. Section II - Rail loading.	(3.6.2)
c. Section III - Containerization.	(3.6.3)

3.6.1 Section I - General. This section shall contain general information for rail transport. The TGTM shall reflect the limitations for CONUS or overseas areas, described in the MTMCTEA Transportability Engineering Analysis on the specific item. It shall identify disassembly of outsize or overweight materiel required to comply with railroad clearance limitations.

3.6.1.1 Safety. Besides the general safety requirements of chapter 1, safety precautions peculiar to rail transport shall be identified in order of severity, including applicable items listed in 3.4.2.1.

3.6.1.2 Material handling. For any piece of equipment that requires handling, the following information shall be provided:

a. Handling procedures identifying materials handling equipment by required type and capacity and recommended slings or sling sets.

b. Handling restrictions and alternate methods applying to the specific item because of sectionalization.

c. Loading ramp strength requirements, angle of incline, method of attaching to railcar, and any special features.

d. Sling arrangements or spreader bars necessary to protect nonstructural areas during lifting operations. The need for spreader bars will be identified in the Transportability Engineering Analysis.

3.6.2 Section II - Rail loading. This section shall provide information on current commercial and STANAG standard loading and tiedown procedures. It shall also provide, when appropriate, information on selecting the appropriate railcar type.

3.6.2.1 Item preparation. This section shall describe item preparation required for rail transport. Guidance presented shall offer the maximum protection from theft and pilferage. Equipment reduction (3.4.3.8) shall be to the extent necessary for rail clearance.

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3.6.2.2 Load restraint. The LRFs used for surface and air modes are the "g" (acceleration of gravity) loading factors expected in military transport. The LRFs used in the rail transport chapter shall be given in a TGTM appendix. Unless otherwise stated in the MTMCTEA Transportability Engineering Analysis, railcar LRFs are the Association of American Railroads (AAR) recommended load restraint values from General Rule Loads:

3.0 in the longitudinal direction (relative to the railcar).

2.0 in the lateral and vertical directions.

The tiedown arrangements shown in the TGTM shall be based on the following:

- restraint factors applied independently in each direction.

- restraint load (GVW times the LRF) resolved into resultant lashing loads, allowing for tiedown angle.

- lashing loads less than the safe working load (SWL) of the restraint (that is, chain or wire rope).

When the equipment has an AAR tested and approved tiedown figure available in Section No. 6, Rules Governing the Loading of Department of Defense Materiel on Open Top Cars, use that tiedown arrangement in the TGTM. The Section No. 6 tiedown figure may not fully comply with the above load restraint values.

3.6.2.3 Bill of materials. The bill of materials shall list the material nomenclature of each item; a description of the item, with size, grade, type, and the applicable Federal Specification or Standard reference and NSN; and the required linear footage to restrain the item. The bill of materials for rail tiedown should be in the same general format used for highway (figure 5).

3.6.2.4 Blocking and tiedown procedures. A blocking and tiedown diagram for each item shall present a loading arrangement using commonly available, commercial restraint devices and materials that will be accessible to Army units. The blocking and tiedown arrangement must provide adequate rail loading restraint (3.6.2.2). Lifting and tiedown provisions shall retain the designation indicated in chapter 1 (3.4.3.7). A table of instructions outlining blocking and tiedown procedures shall accompany each blocking and tiedown diagram. Figure 7 shows sample format and content. The table shall identify the quantity required and steps necessary to ensure proper installation. Where general instructions apply, they shall be listed at the bottom of the table.

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3.6.2.5 Railcar requirements. When an item requires transport by specialized or specific rail equipment, the special equipment characteristics or the type of car required shall be identified.

a. Platcars shall be identified by load capacity, length, width, and platform height.

b. Boxcars shall be identified by load capacity, inside length, width, height, and door opening.

c. Cars with special features, such as chain-equipped, cushioned draft gear-equipped, TOFC hitch, or multilevel (used for trailers and small vehicles), shall be identified to assure compatibility with the item being transported.

3.6.3 Section III - Containerization. When applicable, this section shall provide procedures for the use of MILVANs and commercial containers. Procedures shall include item preparation and required blocking, bracing, and tiedown.

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3.7 CHAPTER 4 - Marine transport. This chapter shall provide procedures for safe, economic movement by breakbulk, roll-on/roll-off (RORO), and when applicable, container ships. Unless otherwise specified (see 6.4), this chapter will not include barge-carrying offshore-discharge ships, Army or Navy lighterage, or other logistics-over-the-shore (LOTS) small craft. This chapter shall include the following:

	Para No.
a. Section I - General.	(3.7.1)
b. Section II - Shiploading.	(3.7.3)
c. Section III - Containerization.	(3.7.4)

3.7.1 Section I - General. This section shall contain general guidance on preparation, handling, and intransit stowage of an item moving through marine terminals. The TGTM shall reflect the limitations for CONUS or overseas areas, described in the MTMCTEA Transportability Engineering Analysis on the specific item.

3.7.2 Safety. Besides the general safety requirements of chapter 1, the TGTM shall identify specific safety precautions pertinent to terminal and marine shipment including applicable items from 3.4.2.1. When applicable, reference shall be made to pertinent DOD and DOT regulations.

3.7.3 Section II - Shiploading. This section shall provide guidance on shiploading procedures. It shall present details on materials required, the item specific loading procedure, and tiedown and blocking.

3.7.3.1 Item preparation. When the item requires specific preparation for marine transport, this section shall provide details. Reduction of all equipment shall be (3.4.3.8) to the extent necessary for hatch or hold clearance, tight stow, and therefore, shipping economy.

3.7.3.2 Security and preservation. Security and preservation of the materiel shall be in accordance with applicable regulations and directives relative to expected terminal environments and types of stowage available onboard ship. Cargo holds offer the maximum protection, while ondeck loading offers minimum protection from the elements and sea conditions. Guidance shall identify any handling restrictions or preparation, such as requirements for peeling back a portion of the closure kit or protective cover for access to lifting or tiedown provisions during the loading/unloading process. Where applicable, alternate procedures shall be described. Guidance presented shall provide reasonable theft and pilferage protection.

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3.7.3.3 Material handling. Handling procedures shall identify the following:

a. Capacities required for shoreside or ship-mounted cranes.

b. A slinging diagram using the MTMC-recommended Navy sling (see MIL-S-22824) set shall be prepared for each item. This diagram (figure 8) shall show attachment points (3.4.3.7) that give a near-level lift (+/-15°) using MTMC-recommended Navy slings. This diagram shall also show the sling lengths necessary for a level lift (+/-3°).

c. Sling arrangements or spreader bars necessary to protect nonstructural areas during lifting operations. The need for spreader bars, when applicable, will be identified in the Transportability Engineering Analysis.

d. When the TGTM item addresses specific ship types (see 6.4), additional information shall be required.

(1) Vehicles that can be ramp loaded onto vessels (barges or RORO ships), under their own power with a towed load, shall be described with applicable notes or cautions. Critical clearances aboard a specific type of vessel that are probable problem areas shall be described.

(2) Nonstandard practices shall be identified; for example, double-boom rigging of an item larger than the hatch opening, requiring one end to be lowered first and swung under an upper deck to allow the other end to be lowered through the hatch.

3.7.3.4 Load restraint. The LRFs used for surface and air modes are the "g" (acceleration of gravity) loading factors expected in military transport. The LRFs used in the marine transport chapter shall be given in a TGTM appendix. Unless otherwise stated in the MTMCTEA Transportability Engineering Analysis, marine LRFs are:

1.2 in the lateral direction (relative to the ship).
0.7 in the longitudinal direction.
0.2 in the vertical direction.

The tiedown arrangements shown in the TGTM shall be based on the following:

- restraint factors applied independently in each direction.

- restraint loads (GVW times the LRF) resolved into resultant lashing loads, allowing for tiedown angle.

- lashing loads less than the safe working load (SWL) of the restraint (that is, chain or wire rope).

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3.7.3.5 Bill of materials. The bill of materials shall list the material nomenclature of each item; the description of the item, with size, grade, type, the applicable Federal Specification or Standard reference, and NSN; and the required linear footage to restrain the item.

3.7.3.6 Blocking and tiedown procedures. The marine mode tiedown procedures for each item shall present a loading arrangement using, where feasible, commercially available restraint devices. The tiedown arrangement must provide adequate marine loading restraint (3.7.3.4). A blocking diagram shall be presented to show the restraint method used when shipboard tiedown points are unavailable. Lifting and tiedown provisions shall retain the designation shown in chapter 1 (3.4.3.7). A table of instructions outlining blocking and tiedown procedures shall accompany each blocking and tiedown diagram. The table shall identify the quantity required and steps necessary to ensure proper installation. Where general instructions apply, they shall be listed at the bottom of the table. Figure 9 shows sample format and content. As noted above, wooden blocking is used only as an alternative to standard shipboard lashing gear. To emphasis this, the marine tiedown figure should include the following statement:

NOTE: Blocking is required only when adequate lashing is unavailable.

3.7.3.7 Alternate methods. At ports where normal stevedoring procedures result in a satisfactory stow, portions of the guidance offered in this chapter may be omitted. Chapter 4 shall, therefore, contain as a NOTE the following statement:

NOTE
The methods described in this chapter for lifting and securing vehicles are recommended procedures. Other methods of handling and stowage may be used provided they ensure safe delivery without damage.

Rationale for the note is that procedures currently used under contracts with stevedoring labor organizations at commercial terminals and outports may be continued with less economic impact than would be incurred by modifying contracts to include specific procedures described in the TGTM.

3.7.4 Section III - Containerization. When applicable, this section shall provide procedures for the use of MILVANs and commercial containers and flatracks. Procedures shall include item preparation and marine mode tiedown, blocking, and bracing.

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3.8 CHAPTER 5 - Air transport. This chapter shall consist of the following:

	Para No.
a. Section I - General.	(3.8.1)
b. Section II - Transport by Cargo Aircraft.	(3.8.2)

3.8.1 Section I - General. This section shall contain information that applies to all aircraft suitable for transporting the particular item. Significant technical details that could influence or limit air transport shall also be presented. When appropriate, this section shall also contain an estimate on manpower, materials (quantity and type), and time needed to prepare, load, tie down, and unload the item from military or commercial aircraft.

3.8.1.1 Safety. Include specific safety precautions peculiar to air transport that are in addition to the general safety requirements of chapter 1. If practical, identify precautions in order of severity, with the greatest hazard potential first and the least hazard potential last.

3.8.1.2 Hazardous material. When military aircraft transport hazardous materials, the provisions of TM 38-250 shall be followed. To alert the shipper, the TGTM shall contain a statement such as the following:

"Written notification is required before shipment of dangerous or hazardous materials aboard military or DOD contract aircraft. Details are in TM 38-250/AFR 71-4/NAVSUP PUB 505(Rev)/MCO P4030.19D/DSAM 4145.3, Preparation of Hazardous Materials for Military Air Shipment."

3.8.2 Section II - Transport by cargo aircraft. This section shall cover the procedures used for loading and tying down equipment aboard fixed-wing cargo aircraft. It shall also identify item preloading preparation required to ensure compatibility with the aircraft type and phase of airborne operation. When the item for transport is a near maximum load for the particular aircraft, this section shall contain a statement indicating whether the aircraft may transport additional cargo and/or personnel.

3.8.2.1 USAF cargo aircraft. This section shall identify any specific item preparation required for USAF cargo aircraft. The prime mission cargo aircraft considered in this section are the C-130, C-141, C-5, and C-17.

3.8.2.2 Tiedown diagrams. When required (see 6.4), the TGTM shall include plan view diagrams showing aircraft tiedown locations and the item tiedown fittings to be used. Refer to MIL-STD-1791 for aircraft details. The legend block for each diagram shall show aircraft type, item nomenclature, direction the item faces in relation to the aircraft, type and rated capacity of aircraft restraint device, and the center of

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gravity for the loaded equipment. Figure 10 shows typical information and layout. The number of restraint devices shown shall be the minimum required for each type of aircraft. Aircraft restraint devices shall not be shown attached to an aircraft fitting of lesser rated capacity. Each tiedown provision shall retain the designation shown in chapter 1 (3.4.3.7). The quantity and type of shoring materials (see 3.8.2.5) required for loading or supporting the item in flight shall be included on the individual aircraft tiedown diagram.

3.8.2.3 Civil Reserve Air Fleet. When required (see 6.4), specific CRAF aircraft shall be covered as described in 3.8.2.1 and 3.8.2.2.

3.8.2.4 Load restraint. The total load restraint capability of the TGTM tiedown arrangement shall be computed and described for each of primary directions of applied force. Current Air Force "g" factors listed in MIL-STD-1791 are 3.0 g's forward, 1.5 g's aft and lateral, 2.0 g's (1.0 g effective) up, and 4.5 g's down. The longitudinal, lateral, and vertical component loads are based on the angles of attachment to the cargo.

3.8.2.5 Shoring. Shoring is required any time the contact pressure of the item being loaded approaches the rated unit floor loading capacity of the aircraft. The shoring shall be described in this section. Plans for concentrated and distributed loads shall be prepared on an individual basis to assure that the floor loading limits established by the manufacturer and/or applicable specifications for the particular aircraft are not exceeded. Bridging shoring will be designed to support 4.5 g's.

3.8.2.6 Restrictions. The TGTM shall identify flight restrictions for hazardous materials that require immediate ejection when preestablished conditions or limits are reached. These restrictions normally include, but are not limited to, a maximum altitude and airspeed for extraction or jettison.

3.8.2.7 Helicopter. When the MTMCTEA Transportability Engineering Analysis addresses an external helicopter transport requirement, the TGTM shall contain the following statement: "Helicopter External Air Transport (EAT) procedures are in FM 55-450-1, Army Helicopter External Load Operations."

3.9 References and Appendixes.

3.9.1 References. References shall be used only to help users better understand the subject. When only a small portion of a document applies, that portion shall be included rather than referenced. References to relevant portions of documents listed in section 2 shall be by subject matter and paragraph heading. References to other manuals shall state the publication number and applicable paragraph or subparagraph number. Each referenced document shall be listed in the TGTM reference appendix.

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3.9.2 Appendixes. A TGTM appendix shall be used to list the transport mode LRFs (3.5.4.5, 3.6.2.2, 3.7.3.4, and 3.8.2.4). An example is given in appendix B. Appendixes are also used when it is necessary or helpful to the reader to furnish additional or supplemental material not a part of the main body of the publication; for example, a list of related reading material (references), glossaries or conversion tables to the metric system of measurement, and a method of estimating tiedowns lashing requirements. (An example is in appendix B.) Appendixes will be placed at the end of the text. Lettering, headings, and page numbering shall be in accordance with MIL-M-38784.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.4), the contractor performs all inspections (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use its or any other facility suitable for the performing the inspections specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in this specification, where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract.

4.2 Final approval. All TGTMs shall be coordinated with and approved by Military Traffic Management Command Transportation Engineering Agency. The address is Commander, MTMC Transportation Engineering Agency, ATTN: MTTE-TR, Box 6276, Newport News, VA 23606-0276.

5. PACKAGING.

5.1 General. The requirements of MIL-M-38784 shall apply.

6. NOTES. (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. TGTMs prepared in accordance with the format requirements of this specification are for use by military troops in shipment of military equipment by all transport modes worldwide.

6.2 General. The provisions of paragraph 6, "Notes," of MIL-M-38784 shall apply. Where references to other paragraphs of MIL-M-38784 conflict with this specification, the provisions of this specification shall apply.

6.3 Technical manual acquisition. This specification must be listed on the Contract Data Requirements List (DD Form 1423) to acquire the technical manuals described by this specification, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

6.4 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced in section 2.

c. Title and publication number of the TGTM to be prepared.

d. Description of the target audience and the required reading grade level.

e. Further tailoring of this specification as provided by a completed content/format selection summary sheet, appendix A.

6.5 Subject term (key word) listing.

Air transport
Blocking
Cargo tiedowns
Containerization
Curb weight
Highway transport
Lifting
Marine transport
Rail transport
Reduced
Sectionalized
Tiedown
Transportability

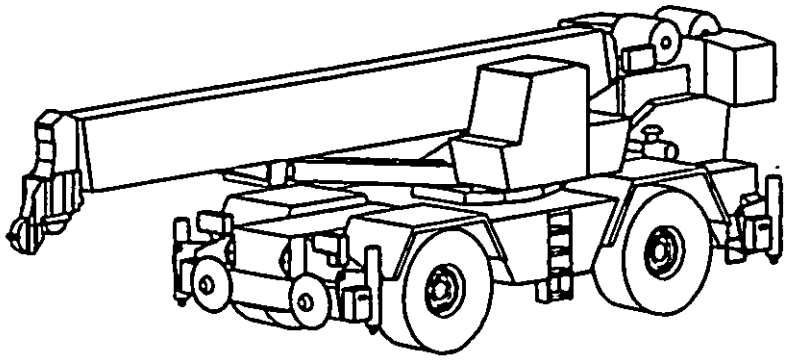
6.6 Additional information. When transportation requirements present problems not covered by this or referenced specifications, additional information may be obtained from the proponent agency or MTMCTEA.

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Publication Number ——— TM 55-3810-306-15

TRANSPORT GUIDANCE TECHNICAL MANUAL

Line Drawing ———



Nomenclature ———

National Stock Number (NSN) (use back of Front Cover for NSN lists) ———

ROUGH TERRAIN CONTAINER CRANE (RTCC)

(NSN 3810-01-205-2716)

See MIL-M-38784 for proper Distribution Statement ———

Approved for public release; distribution is unlimited.

HEADQUARTERS, DEPARTMENT OF THE ARMY

January 1991

MONTH and YEAR ONLY

FIGURE 1. Title Page (3.3.1)

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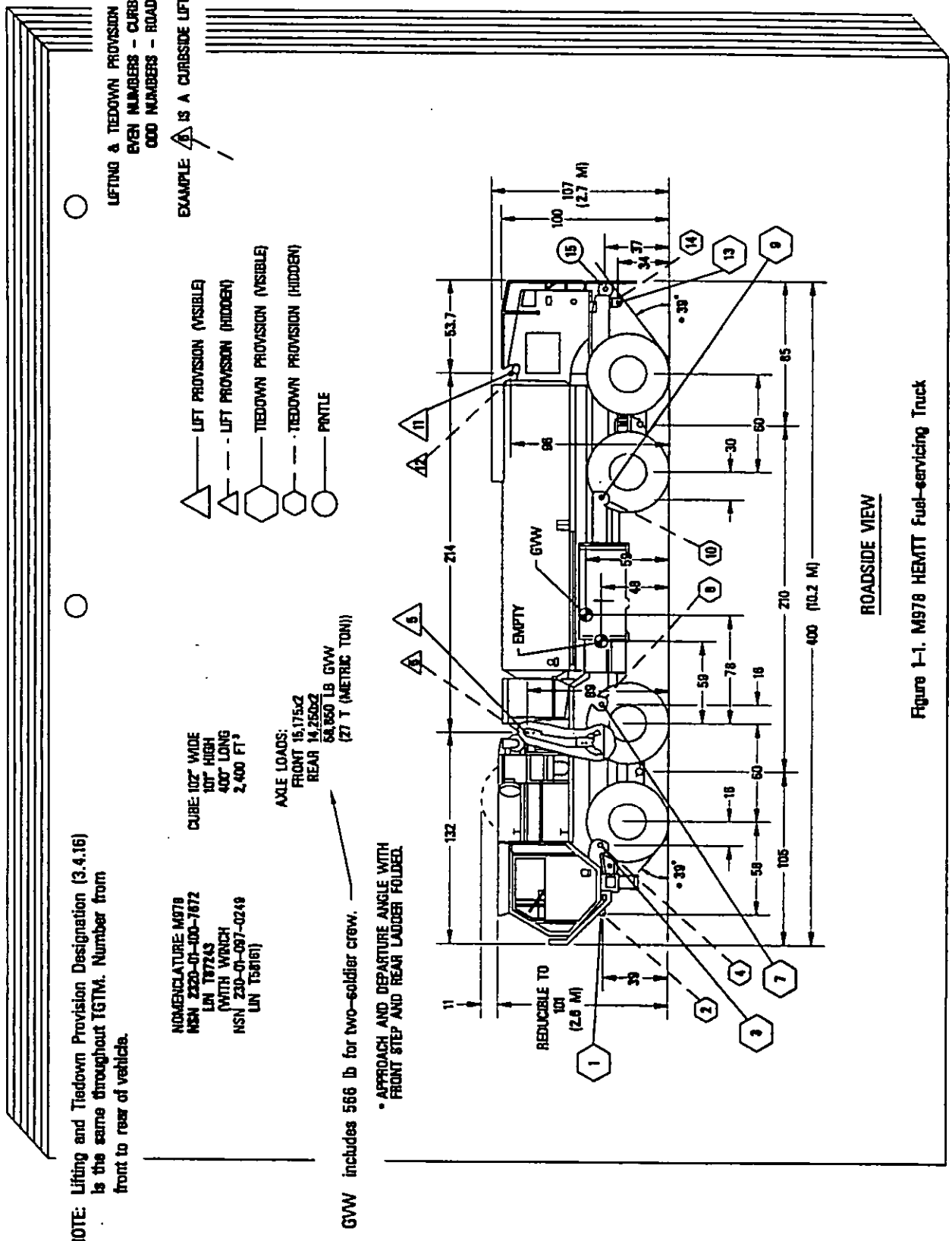


FIGURE 2. Technical Drawing (3.4.3.6)

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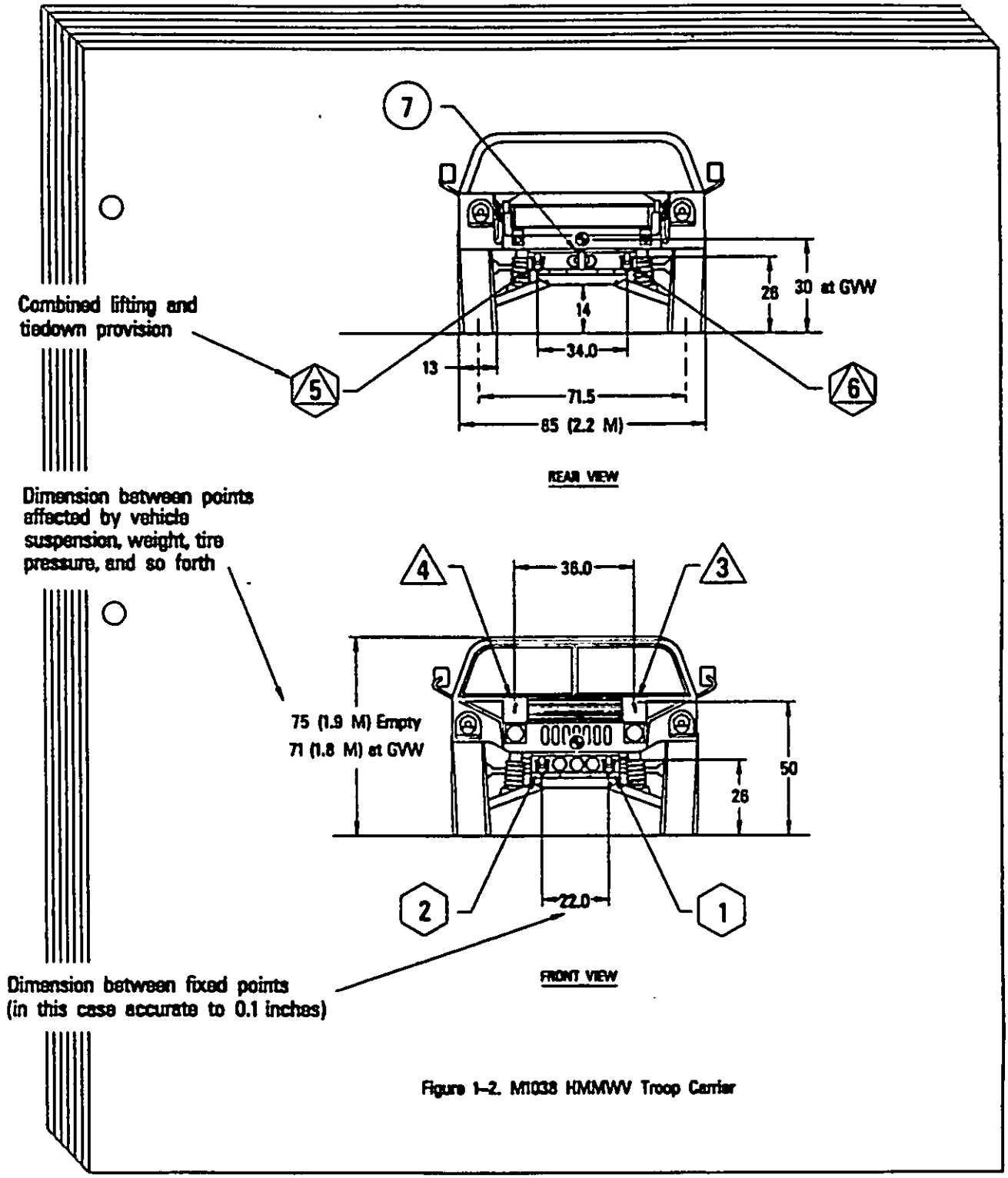


Figure 1-2. M1038 HMMWV Troop Carrier

FIGURE 3. Technical Drawing (3.4.3.11)

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Section II. MOTOR VEHICLE TRANSPORT

2-3. Prime Mover Selection

The physical size and weight of the 6K VRRTFL allow highway transport of the forklift by a variety of vehicles. In selecting a transporter, factors such as the proposed route and local availability of wide load permits must be considered.

Table 2-1. Evaluation of Prime Movers

Semitrailers	Load Rating (tons)	Comments
M747 HET	60	Much too large.
M870 Lowbed	40	Larger than needed, but usable.
M872 Flatbed	34	Not well suited. Larger than needed, but usable. Deck height is 58 in.
M172A1 Lowbed	25	Best. Deck is 115 in. wide and 40 in. high.
M871 Flatbed	22.5	Usable. Deck is 96 in. wide by 55 in. high. Slight overhang of 3 in. per side.
M127A2C Stake	12	Not well suited. Deck height is 57 in. VRRTFL counterweight must be removed and shipped separately.
M345 Flatbed	10	Payload too small.

When properly loaded on an M172A1 or M871, the 6K VRRTFL will not overload the transporter or exceed axle limits in most geographic areas.

2-4. Preparation

Preparation for highway transport includes:

a. Filling fuel tank one-quarter of its capacity.
b. Removing all trash and mud from the VRRTFL.
c. Stowing MLRS pod lifting tool on the machine.
d. Checking for fluid leaks and repairing any defects.
e. Securing battery.
f. Ensuring that engine, brakes, and tiedowns are in good mechanical condition.
g. If required, removing the 2,900-pound counterweight to reduce the total weight of the forklift (see Operator's Manual).

FIGURE 4. Vehicle Selection (3.5.4.1)

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2-5. Loading Procedures

a. Material. Table 2-2 shows the bill of materials for blocking and tiedown on a flatbed trailer.

Table 2-2. Bill of Materials for Transport by Semitrailer

Item	Quantity	Description
Chain	8	Chain Assemblies, 3/8-in. by 10-ft High-test welded steel, 8,250-pound Safe Working Load (16,500-pound breaking strength); equipped with matching grabhooks NSN 4010-00-803-8858
Load binders	8	Heavy-duty, double grabhook, Type 4, 3/8-1/2-in. (26,000-pound breaking strength) NSN 3990-01-213-1746

NOTE

Chain assemblies and load binders are Basic Issue Items (BII) for the M172 trailer.

Load binders are generally marked with an ultimate breaking strength rating. Depending on manufacturer, breaking strength is about 3 times the safe working load.

Chain is generally rated by proof test load or about 2 times the safe working load.

b. Loading. Place the forklift in the tiedown position on the semitrailer using a crane of adequate capacity (see para 4-4 for lifting guidance), or drive the forklift onto a semitrailer if a suitable ramp is available. Position the forklift so its weight is distributed relatively equal over the tractor and trailer axles. Set the parking brake. Place the transmission in neutral. Lower the forks to rest on trailer deck.

c. Tiedown. See Figure 2-1 and table 2-3 for instructions on restraining the forklift against forces encountered at normal speeds and operating conditions. Wheel blocking of the VRRTFL is not required. The highway loading restraint factors (LRF) used in Figure 2-1 are listed in Appendix B.

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FIGURE 5. Materials Required for Blocking and Tiedown (3.5.4.6)

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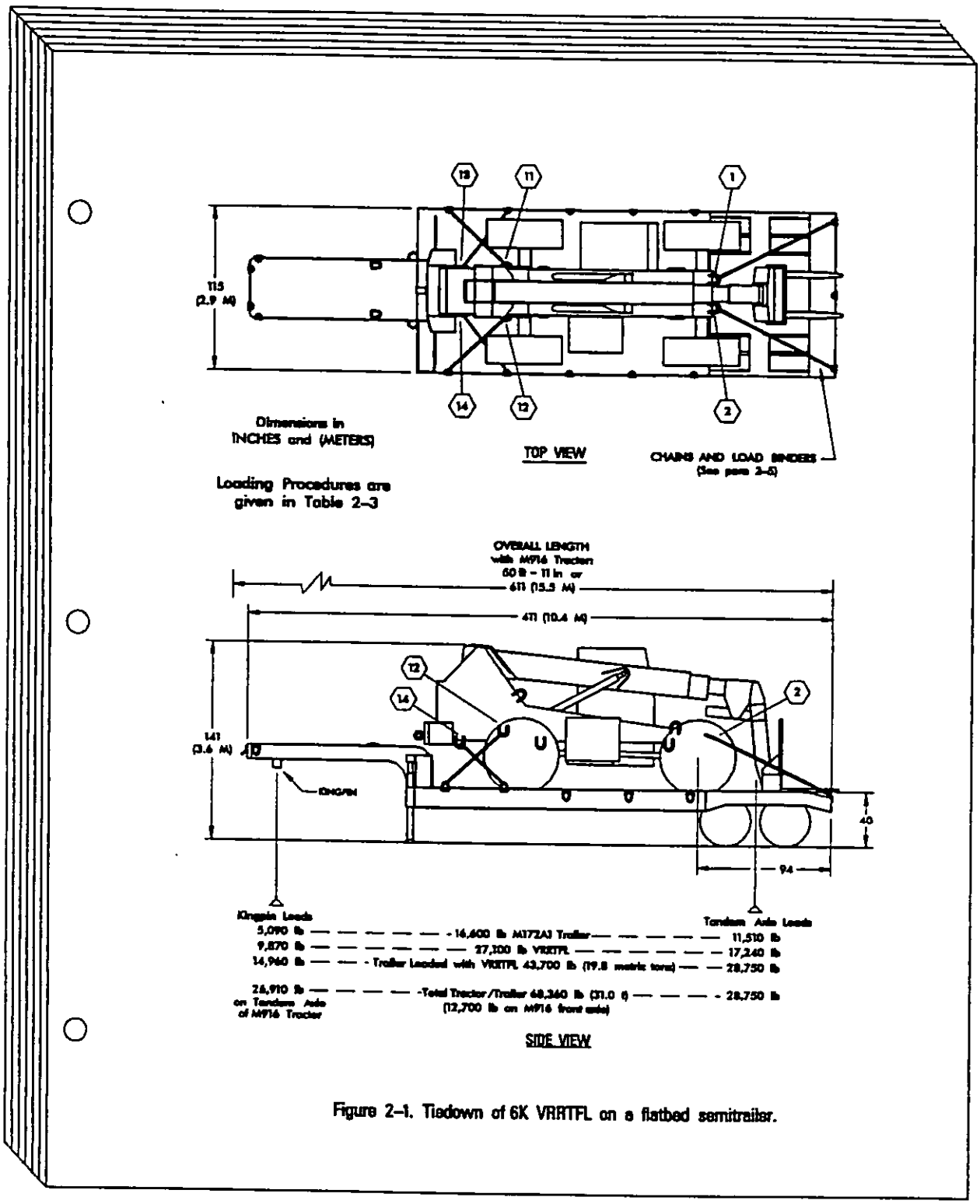


Figure 2-1. Tiedown of 6K VRRTFL on a flatbed semitrailer.

FIGURE 6. Blocking and Tiedown Procedures (3.5.4.7)

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Table 1-1. Tiedown Procedures for Securing 6K VRRTFL on Flatcar (Fig 1-1)

Item	Procedure
Chain tiedowns	Install at indicated locations. The number of chains used will depend on chain size/strength. Use eight 1/2-in. chains (4 in each direction), or twelve 3/8-in. chains (6 in each direction).
	NOTE When chain tiedown-equipped railcars are not available, wire rope and wooden blocking must be used.
Wire rope loops	Each tiedown is made from one piece of 1/2-in. wire rope (8 required), or 5/8-in. wire rope (4 required). Length as required (12 to 16 feet). Form a complete loop between forklift-truck tiedown provision and railcar stake pocket. The angle between the cables and railcar deck (as viewed from the side) should be as close to 45° as possible. The wire rope ends must overlap at least 24 inches. Tension the wire rope by using two cable grippers and a (3-ton) chain-hoist or "come-along."
Cable clamps	Space clamps 3-3/4-in. apart as shown. With 1/2-in. wire rope, use 1/2-in. cable clamps and alternately torque nuts to 65 ft-lb. (With 5/8-in. wire rope, use 5/8-in. cable clamps torqued to 95 ft-lb.)
Thimbles	Use thimbles to protect cable at sharp bends. Attach to cable with cable clamps (5/8-in. clamps on 1/2-in. thimbles and 3/4-in. clamps on 5/8-in. thimbles). Thimbles are optional for the VRRTFL fittings.
Blocking	Position and nail blocking to railcar.
Strapping	Band operators' cab to keep door closed in-transit.

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FIGURE 7. Blocking and Tiedown Procedures Table (3.6.2.4)

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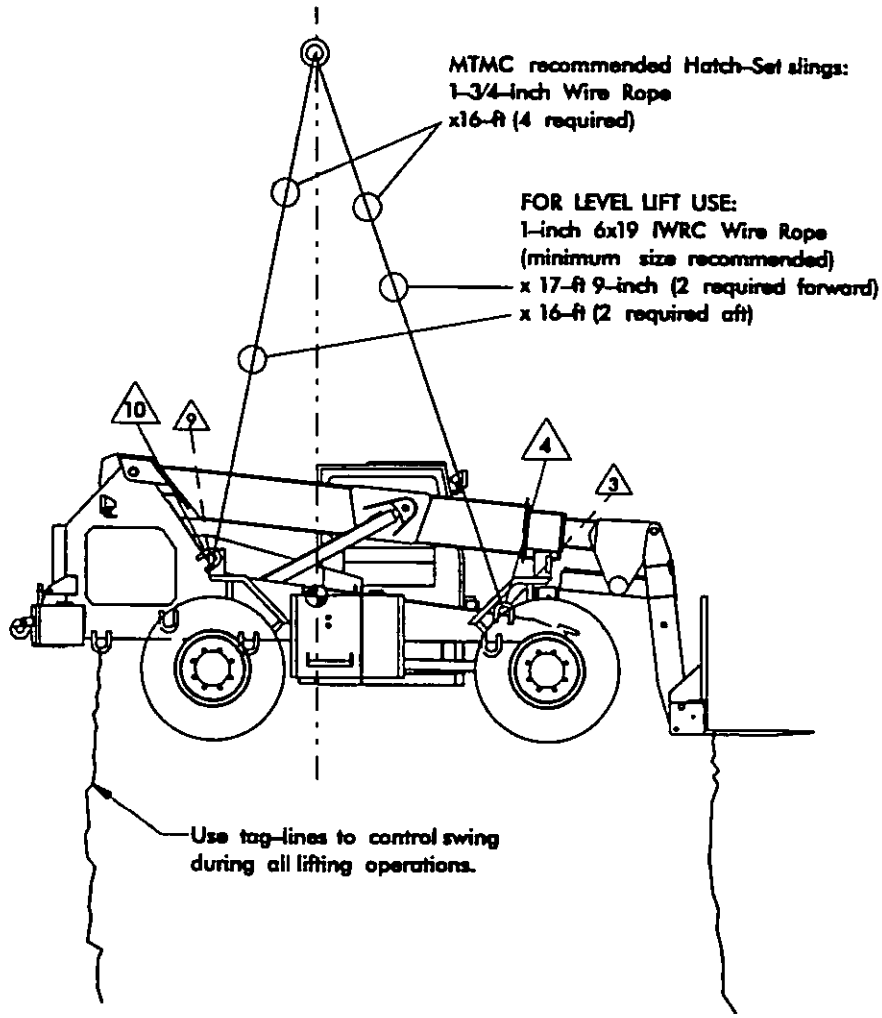


Figure 4-2. Crane lifting of the 6K VRRTFL.

FIGURE 8. Slinging Diagram (3.7.3.3)

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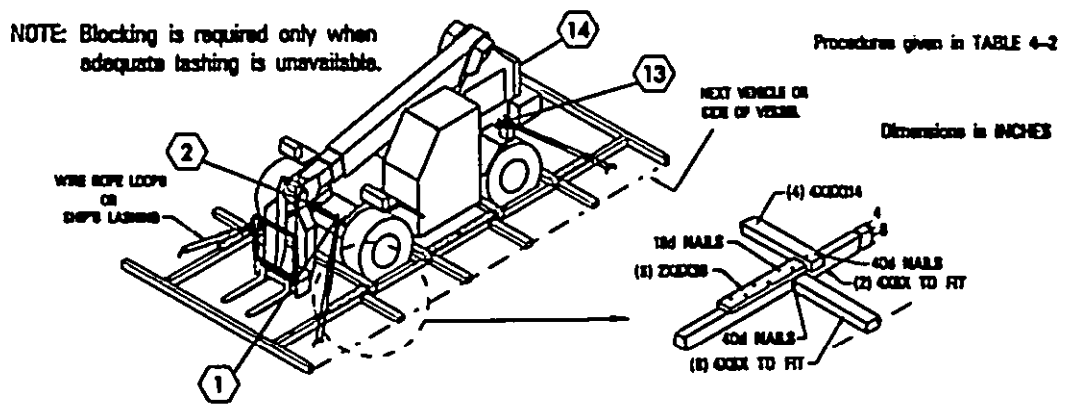


Figure 4-1. Tiedown in General Cargo Ship

Table 4-2. Tiedown of VRRTFL in the Hold of a General Cargo Ship
(see Figure 4-1)

Item	Procedure
Wire rope loops (4 required)	Each tiedown is made from one piece of 1/2 -in. wire rope. Length as required (12-16 ft). Form a complete loop between VRRTFL and ship tiedown provisions. The angle between the cables and deck (as viewed from the side) should be as close to 45° as possible. The wire rope ends must overlap at least 24 inches. Tension the wire rope by using two cable grippers and a (3-ton) chain hoist or "come-along."
Cable clamps	Space clamps 3-3 /4 inches apart. Use 1/2-in. cable clamps and alternately torque nuts to 65 ft-lb.
Thimbles	Use thimbles to protect cable at sharp bends. Attach to cable with 5 / 8-in. cable clamp.
Blocking	Position and nail blocking.

NOTE

The methods described in this chapter for lifting and securing vehicles are recommended procedures. Other methods of handling and stowage may be used provided they ensure safe delivery without damage.

FIGURE 9. Marine Blocking and Tiedown Procedures (3.7.3.6)

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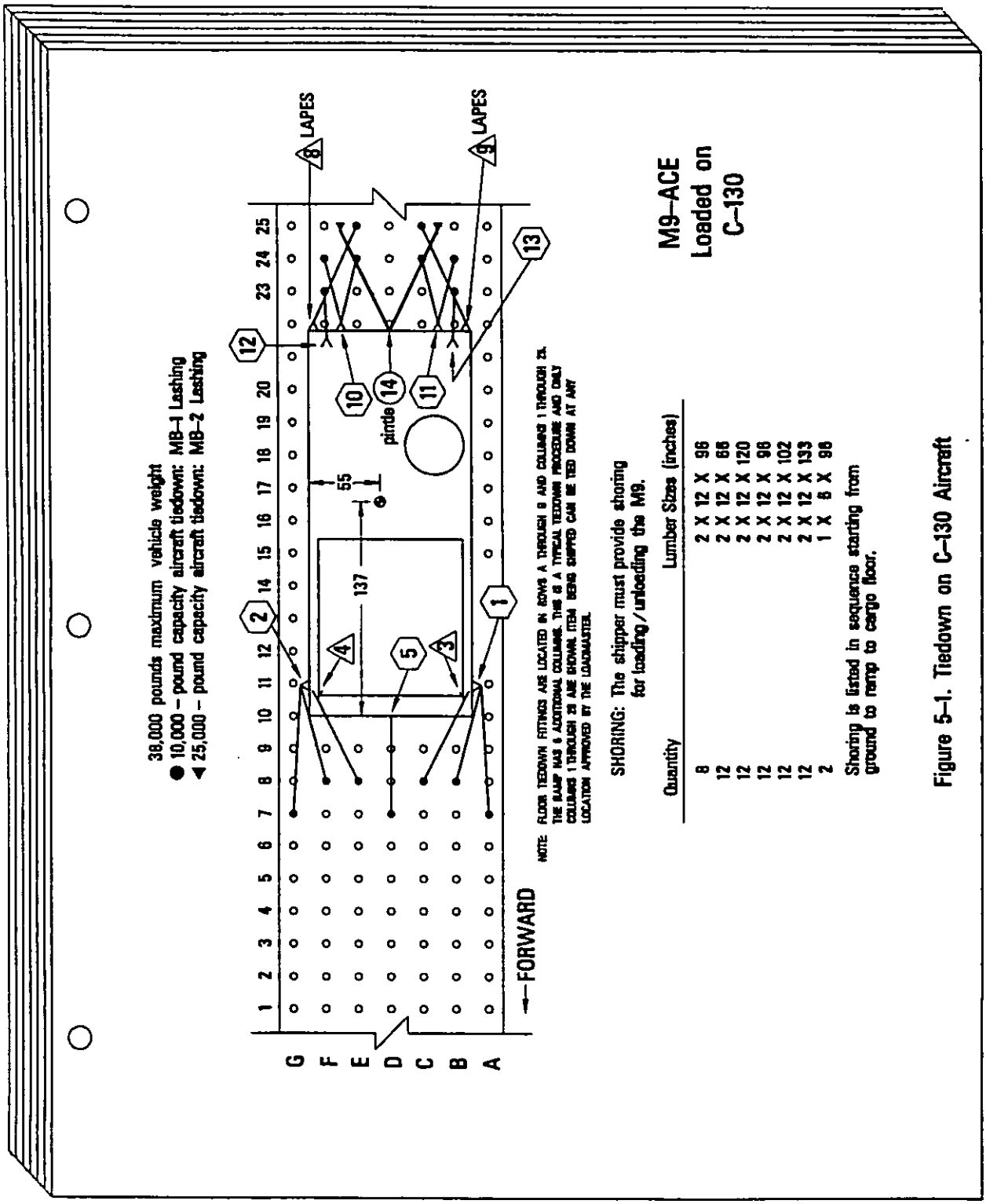


Figure 5-1. Tiedown on C-130 Aircraft

FIGURE 10. Aircraft Tiedown Diagram (3.8.2.2)

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APPENDIX A.

CONTENT/FORMAT SELECTION SUMMARY

Transport Guidance Technical Manual (TGTM)

10. SCOPE

10.1 Scope. The contracting activity will use this appendix to specify which optional requirements of this specification are to be contractually imposed in the acquisition of a Transport Guidance Technical Manual (TGTM). This appendix is a mandatory part of this specification, and the information contained herein is for compliance.

10.2 Application. This appendix is intended to be copied/ reproduced, completed, signed, and become part of the TGTM Contract Document Summary List for solicitation/contract application. A separate content/format selection summary sheet is required for each TGTM acquisition.

10.3 Explanation of columns - Content/Format Selection Summary.

a. Column (1) Item Number - Self explanatory

b. Column (2) Requirement - Identifies the requirement

c. Column (3) Applicable Paragraph Number - Identifies the paragraph where the "Requirement" is described.

d. Column (4) Requirement Selection - Mark either (YES) or (NO)

e. Column (5) Explanation/Remarks - Shall be used when a YES or NO is not adequate to completely identify the requirement. If necessary, additional information may be attached to the Content/ Format Selection Summary sheet.

10.4 Tailoring requirements. Each requirement listed on the Content/Format Selection Summary sheet must be marked YES or NO, with explanation/remarks added as necessary, to assure that selected requirements are adequately identified. Reference to the contract statement of work (SOW) or other contract documents/ attachments containing applicable information may be made in the Explanation/Remarks column.

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Equipment name/nomenclature _____

(1)	(2)	(3)	(4)		(5)
Item No.	Requirement	Applicable Paragraph Number	Requirement Selection YES	NO	Explanation/ Remarks
1	Reading Grade Level	3.1.2			
2	Foldout pages	3.1.4.2			
3	Content	3.2			
4	Warning Page	3.3.2			
5	Units of Measurement	3.4.3.9			
6	Container-ization	3.4.4			
7	Turning Clearance	3.5.3.2b			
8	Ship types	3.7 & 3.7.3.3d			
9	Tiedown Diagrams	3.8.2.2			
10	Civil Reserve Air Fleet	3.8.2.3			
11	Inspection	4.1			

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APPENDIX B

PART 1:
Example of a Loading Restraint Factors appendix. This appendix was used in the 6,000-pound capacity variable reach rough-terrain forklift truck (6K VRRTFL) TGTM.

PART 2:
Example of an appendix showing a method of estimating tiedown lashing requirements. This appendix was also used in the 6K VRRTFL TGTM.

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PART 1: EXAMPLE APPENDIX

Loading Restraint Factors

The loading restraint factors (LRFs) used for surface and air modes are the "g" (acceleration of gravity) loading factors expected in military transport. The tiedown arrangements shown in the 6K VRRTFL TGTM are based on the following:

- restraint factors applied independently in each direction.
- restraint loads (GVW times the LRF) resolved into resultant lashing loads, allowing for tiedown angle.
- lashing loads less than the safe working load (SWL) of the restraint (that is, wire rope).

Highway:

The Transportation Engineering Agency highway LRFs are:
0.7 in the forward direction (relative to the transporter).
0.3 in the aft and vertical directions.
0.1 in the lateral direction.

Rail:

The Association of American Railroads recommended rail LRFs are:
3.0 in the longitudinal direction (relative to the railcar).
2.0 in the lateral and vertical directions.

Marine:

The Military Sealift Command (MSC) design LRFs are:
1.2 in the lateral direction (relative to the ship).
0.7 in the longitudinal directions.
0.2 in the vertical direction.
MSC LRF's are for severe conditions.
Actual marine LRF's vary. Marine tiedown restraint depends on the ship's size (decreasing on larger vessels), the expected sea state, and the stow location on a given ship. Generally, the restraint required will increase for locations high and forward (or aft) in the ship. The most severe conditions occur on exposed "weather-decks," where strong wind and wet conditions add to the problem. A ship's crew may require additional lashing on exposed decks. Below-deck locations that are closer to the vessel's centers of gravity and rotation will experience less severe motion. The blocking and tiedown procedures presented (figure 4-1) allow for severe, below-deck conditions.

Air:

The USAF aircraft LRF's are:
3.0 in the forward direction (relative to the aircraft)
2.0 in the vertical direction.
1.5 in the aft and lateral directions.

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PART 2: EXAMPLE APPENDIX

Estimating Tiedown Lashing Requirements

The number of lashings required to safely tie down the VRRTFL for highway transport on a typical truck/semitrailer can be estimated as follows:

STEP 1 - Determine the amount of longitudinal restraint needed. [With this method, the required vertical and lateral restraint is covered by the longitudinal factor or factors.]

- The highway loading restraint factor (LRF) is 0.7 (from appendix B) in the forward direction:
  27,100 (VRRTFL GVW) x 0.7 = 18,970 pounds

- The highway LRF is 0.3 in the aft direction:
  27,100 x 0.3 = 8,130 pounds

STEP 2 - Determine the number of chains needed.

- The angle to semitrailer deck and the angle to the side are assumed to be 45°. [cos 45 x cos 45 = 0.5]

- The SWL of 3/8-inch chain (from table 2-2) is 8,250 pounds:

$$\text{No. of forward loading chains} = \frac{18,970}{8,250 \times 0.5} = 4.6 \text{ (rounded up to 5)}$$

While five satisfies the equation; six chains are needed for a symmetric (side-to-side) tiedown arrangement. Therefore, use six chains to prevent the VRRTFL from moving forward on the semitrailer.

$$\text{No. of aft loading chains} = \frac{8,130}{8,250 \times 0.5} = 1.9 \text{ (rounded up to 2)}$$

Use two chains to prevent the VRRTFL from moving aft on the semitrailer.

NOTE:
This method is more conservative than the method in the highway transport chapter. Fewer chains are used in figure 2-1, "tiedown of 6K VRRTFL on a flatbed semitrailer," because the figure is based on the actual tiedown-lashing angles [angle to semitrailer deck (side view) and the angle to the side (top view)].

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3. DOCUMENT TITLE
Manual, Technical: Transport Guidance for Military Materiel, Preparation of

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